NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-8805/55F
26 February 1993
SUPERSEDING
MIL-S-8805/55E
25 November 1987

# PERFORMANCE SPECIFICATION SHEET

## Ⓕ SWITCH, PUSH, MICROPHONE, DPST, NORMALLY OPEN, FOOT OPERATED, MOMENTARY, UNSEALED AND WATERTIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

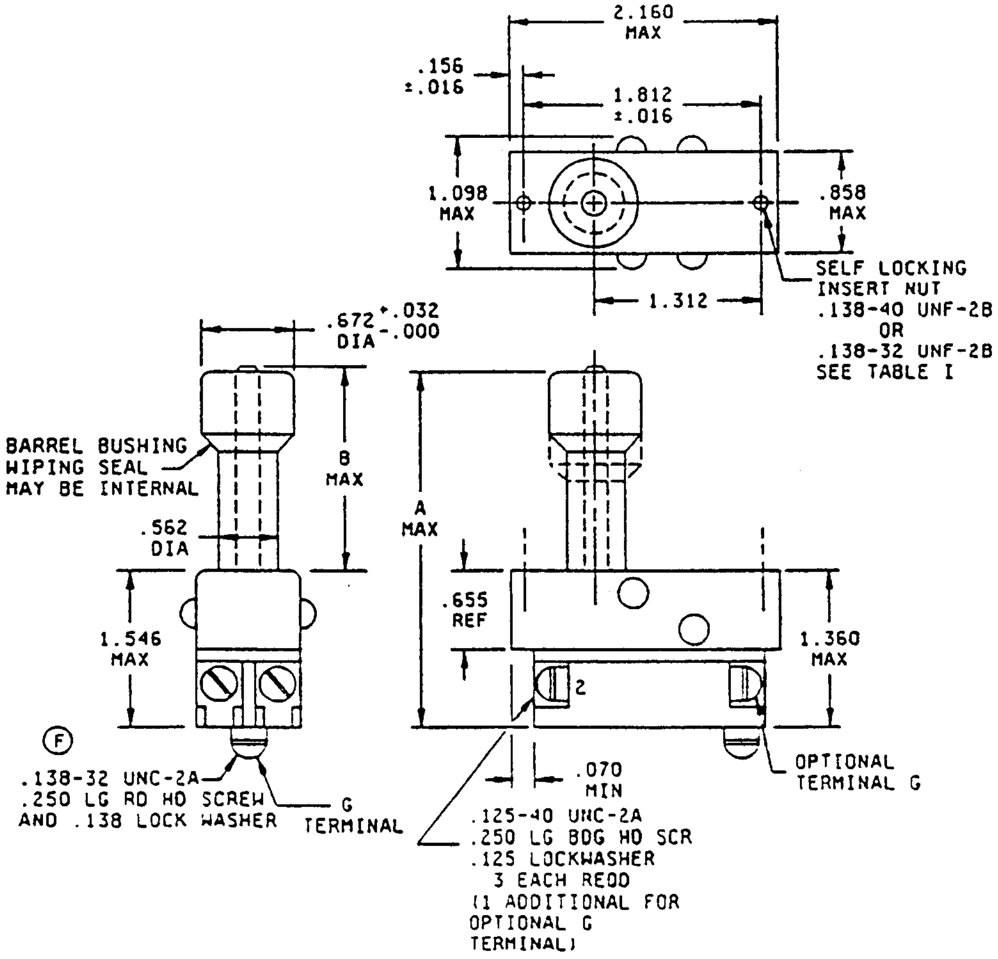

2.160 MAX
.156 ±.016
1.812 ±.016
1.098 MAX
.858 MAX
SELF LOCKING INSERT NUT .138-40 UNF-2B OR .138-32 UNF-2B SEE TABLE I
1.312
.672 +.032 -.000 DIA
BARREL BUSHING WIPING SEAL MAY BE INTERNAL
B MAX
A MAX
.562 DIA
.655 REF
1.546 MAX
1.360 MAX
2
Ⓕ
.138-32 UNC-2A .250 LG RD HD SCREW AND .138 LOCK WASHER
G TERMINAL
.070 MIN
.125-40 UNC-2A .250 LG BDG HD SCR .125 LOCKWASHER 3 EACH REQD (1 ADDITIONAL FOR OPTIONAL G TERMINAL)
OPTIONAL TERMINAL G

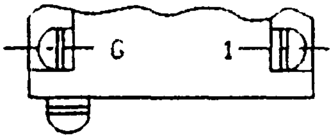
FIGURE 1. Switch, microphone.

Ⓕ denotes changes

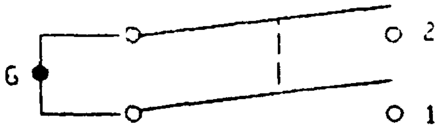
AMSC N/A

FSC 5930

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.



TERMINAL MARKINGS



SCHEMATIC DIAGRAM

| Inches | mm | Inches | mm |
|---|---|---|---|
| 010 | 0 41 | 858 | 21 79 |
| 016 | 0 41 | 910 | 23 11 |
| 032 | 0 81 | 1 098 | 27 89 |
| 070 | 1 78 | 1 312 | 33 32 |
| 125 | 3 18 | 1 360 | 34 54 |
| 138 | 3 51 | 1 546 | 39 27 |
| 156 | 3 96 | 1 660 | 42 16 |
| 250 | 6 35 | 1 812 | 46 02 |
| 562 | 14 27 | 2 160 | 54 86 |
| 655 | 16 64 | 2 218 | 56 34 |
| 672 | 17 07 | 2 968 | 75 39 |

NOTES

1 Dimensions are in inches

2 Unless otherwise specified, tolerance is ± 010

3 Metric equivalents are given for general information only

4 Mounting hardware not furnished

Lockwashers are not captive to terminal screws

6 Dimension A is 2 218 or 2 968 see table I

7 [illegible] see table I

8 Design optional within envelope dimensions shown

[illegible]

REQUIREMENTS

Dimensions and configuration  See figure 1 and table I

Ⓕ Enclosure design  1 (unsealed) or 3 (watertight), see table I

Temperature characteristic  -40°C to +85°C

Thermal shock: In accordance with test condition A, method 107 of MIL-STD-202, except low temperature extreme shall be -40°C, (-40°C to +85°C)

Shock  50 g's sawtooth  in accordance with method 213  test condition G, MIL-STD-202

Sinusoidal vibration grade  1 (10 to 500 Hz)

Ⓕ Material

Mounting bracket  Steel, corrosion-resistant or suitably protected to resist corrosion

Weight

With short actuator bushing (M8805/55-001, -003, -005, and -007)  0 25 pound maximum

With long actuator bushing (M8805/55-002, -004, -006, and -008)  0 32 pound maximum

Operating characteristics

Actuating force  7 0 (+4 -0) pounds

Pretravel  090 inch minimum

Total travel  165 inch minimum.

Overtravel  050 inch minimum

Differential travel  050 inch minimum

Full overtravel force  20 pounds maximum

Release force  0 5 pound minimum

Ⓕ Rated load at 28 V dc  0 75 ampere inductive at sea level only

Overload at 28 V dc

Inductive  1 75 amperes

Minimum operating cycles

Rated load  50,000 cycles

Overload  1,000 cycles

Dielectric withstanding voltage

Altitude  50,000 feet

Terminal strength

Pull  20 pounds

Torque  5 inch pound.

Torque 5 inch-pounds

Strength of actuating means 20 pounds

Ⓕ Moisture resistance Insulation resistance shall be not less than 2 5 megohms in the wet condition, and not less than 50 megohms in the dry condition

Salt spray (corrosion)

Unsealed switches (M8805/55-001 through -004) Group III of the MIL-S-8805 qualification inspection table is applicable, except no electrical tests applicable following salt spray

Sand and dust Method I

Solderability Not applicable

Mechanical endurance 50,000 cycles

Qualification inspection See table II

Part or Identifying Number (PIN) See table I

Ⓕ TABLE I PIN's, configurations and dimensions

| Part or Identifying Number PIN 1/ | Seal | Actuator bushing | Dimensions | | Insert nut thread |
|---|---|---|---|---|---|
| | | | A | B | |
| M8805/55-001<br>M8805/55-002 | Unsealed | Short<br>Long | 2 218<br>2 968 | 0 910<br>1 660 | 138-40 UNF-2B |
| M8805/55-003<br>M8805/55-004 | | Short<br>Long | 2 218<br>2 968 | 0 910<br>1 660 | 138-32 UNF-2B |
| M8805/55-005<br>M8805/55-006 | Watertight | Short<br>Long | 2 218<br>2 968 | 0 910<br>1 660 | 138-40 UNF-2B |
| M8805/55-007<br>M8805/55-008 | | Short<br>Long | 2 218<br>2 968 | 0 910<br>1 660 | 138-32 UNF-2B |

1/ M8805/55-001 replaces and is interchangeable with canceled PIN MS25039-1

Ⓕ TABLE II Qualification inspection group submission

| Examination or test 1/ 2/ | Basic switch | Other switch samples 1/ | Extent of approval |
|---|---|---|---|
| Groups I II, III, VI, VII, and VIII | M8805/55-005 | M8805/55-008 Group I | All |
| Groups I, II, III, and V | M8805/55-001 | | |

1/ Group numbers refer to those inspection groups specified in the MIL-S-8805 qualification inspection table

2/ [illegible] ealed version or only the [illegible] are being qualified [illegible] qualification inspection table of [illegible]

## CONCLUDING MATERIAL

Custodians:
  Army - ER
  Navy - EC
  Air Force - 85

Review activities:
  Army - AR
  Air Force - 99
  DLA - ES

User activities:
  Navy - AS, MC
  Air Force - 19

Preparing activity:
  Navy - EC

Agent:
  DLA - ES

(Project 5930-1482)